

PRODUCT DESCRIPTION

HeatShield is a single-component, waterborne reflective waterproofing coating for roofs, decks and exterior surfaces. The dried film is loaded with hollow ceramic microspheres: rather than letting solar radiation be absorbed into the roof, the spheres reflect it back off the surface, while the air sealed inside them slows conduction of the remaining heat into the structure below. Reflectance testing by the Research Office of King Mongkut's University of Technology Thonburi returned a total solar reflectance above 95%, and in service the coating typically lowers roof surface temperature by 5 °C or more. The film is flexible, water resistant and bridges movement up to approximately 2 mm, so one coating both waterproofs and cools. It bonds to concrete and cement render, metal-sheet and tiled roofing and polycarbonate sheet.

KEY DATA

Chemistry	Single-component waterborne polymer with hollow ceramic microspheres
Supplied form	Ready-to-use liquid — no mixing, no thinning. Brush, roller or airless spray.
Appearance / colour	White only — matt film (gloss < 10 GU). The light colour is what produces the reflectance.
Physical data	52 ± 1 % non-volatile (D1644-01) · SG 1.12 ± 0.01 (D1478) · 3,000 ± 100 cps Brookfield / 90 ± 10 KU · fineness < 50 micron
VOC / flammability	< 10 g/L (D3960) · no flash point detected (D92) — the wet material is not flammable
Solar performance	TSR > 95 % (C1549 / E903-96) · emissivity 0.75 (C1371-04a / E408), tested by the Research Office, KMUTT · surface temperature down 5 °C or more in service
Film properties	Bridges cracks up to approx. 2 mm · pencil hardness > HB · cross-cut adhesion 25 / 25
Drying / system	Touch dry < 3 min · surface dry < 15 min · hard dry < 40 min · 2 coats, the second cross-wise, recoat after 1–2 h · in service approx. 4 h after the final coat
Primer	Bare, previously uncoated concrete: HeatShield Primer first, allow approx. 30 min
Coverage	Approx. 5 m ² per kg over 2 coats (4.5 kg ≈ 22.5 m ² · 18 kg ≈ 90 m ²)
Packaging	HeatShield 4.5 kg and 18 kg · HeatShield Primer 4.5 kg and 18 kg

NOTE ON VALUES

The coverage figure is the standard basis used for conventional paints and is for budgeting rather than exact ordering: deep-profile sheeting, rough surfaces and porous concrete all consume more. Reflectance and emissivity are laboratory results on a prepared panel; in-service values depend on film thickness and on how clean the surface is kept.

APPLICATION NOTES

- Surface preparation: clean, dry and free of dust, oil, grease, efflorescence and loose or flaking paint. On previously painted surfaces remove all chalking or poorly adhering coating down to a sound base.
- Repair before coating: fill cracks, open joints and spalls, seal laps and screw penetrations on sheet roofing. Coating over an unrepaired crack wider than the bridging capability will fail along that line.
- Do not thin with water to increase coverage — reflectance and durability are both functions of film thickness and pigment loading. Clean tools with water immediately, while the material is still wet.
- Do not apply onto standing water or a wet surface, or if rain is expected within about 4 hours. Not below 10 °C.
- Reflective performance depends on the surface staying light in colour: rinse accumulated dirt off periodically. Some loss of reflectance between cleanings is normal for any reflective roof coating.
- Scope: HeatShield is the entry level of the LucernaPro waterproofing range — it sheds water and reduces heat, but is not intended for permanent immersion or standing water. Specify LucernaPro SiliconePro or above for those.

HEALTH & SAFETY

Waterborne, very low in VOC (< 10 g/L) and no flash point detected — the wet material is not flammable. Even so, wear gloves and eye protection, ventilate enclosed areas and rinse splashes with water. The main hazard here is the work at height, not the chemistry: use fall protection and never walk on a roof while the coating is wet and slippery.

STORAGE

Store sealed in the original container at 10–30 °C, dry and out of direct sunlight. Shelf life 12 months unopened.

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The information herein is given in good faith based on our current knowledge and practical experience. Coverage, drying behaviour and in-service reflectance depend on substrate and site conditions; values are typical guidance only and do not constitute a specification. Made in Thailand.